FutureProof EPM

Inflation-Indexed Annuity Impact Analysis

Optimised Model (Principal & Interest)

v14a Optimised — 50,000-Path Monte Carlo Simulation

March 2025

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Parameter	Optimised Value
Model	v14a Optimised (PI)
Mortgage Type	Principal & Interest (amortises to $0)
Property Value	$2,000,000
Initial Mortgage	$1,350,000
Peak Mortgage (Year 10)	$1,600,000
Mortgage at Maturity	$0
Holiday Entry	1.05x (vs 0.90x baseline)
Profit Share	20% every 3 years
Baseline PoC (Year 30)	4.94%
Inflation Rates Tested	2.0%, 2.5%, 3.0%, 3.5%, 4.0%

1. Executive Summary

This analysis examines the impact of inflation-indexing the EPM annuity payout using the **optimised Principal & Interest model**. Unlike the interest-only baseline, this model amortises the mortgage to zero over 20 years (from Year 11 to Year 30), dramatically reducing PoC from ~14% to ~5%. The question is whether this structural improvement provides enough headroom to absorb inflation-indexed payouts.

Key Findings

• The optimised PI model absorbs inflation indexing **significantly better** than the IO model. At 3% inflation, PoC rises by only +0.85pp (from 4.94% to 5.79%), compared with +2.13pp in the IO model.

• Even at 4% inflation, PoC remains at just 6.16% — well below the unoptimised IO baseline of 13.87% *without* inflation indexing.

• Total annuity at 3% inflation increases by $36,597 (14.6%), identical to the IO analysis. But the surplus impact is smaller: $48,700 vs $81K in IO.

• The cost multiplier drops to ~1.3x (from ~2.2x in IO), because PI amortisation reduces the mortgage balance faster, limiting the compounding of interest and margin costs.

• P5 surplus shifts from $8,572 (positive) to -$99,528 (negative) — the tail risk boundary is more sensitive, but the absolute level remains far better than IO.

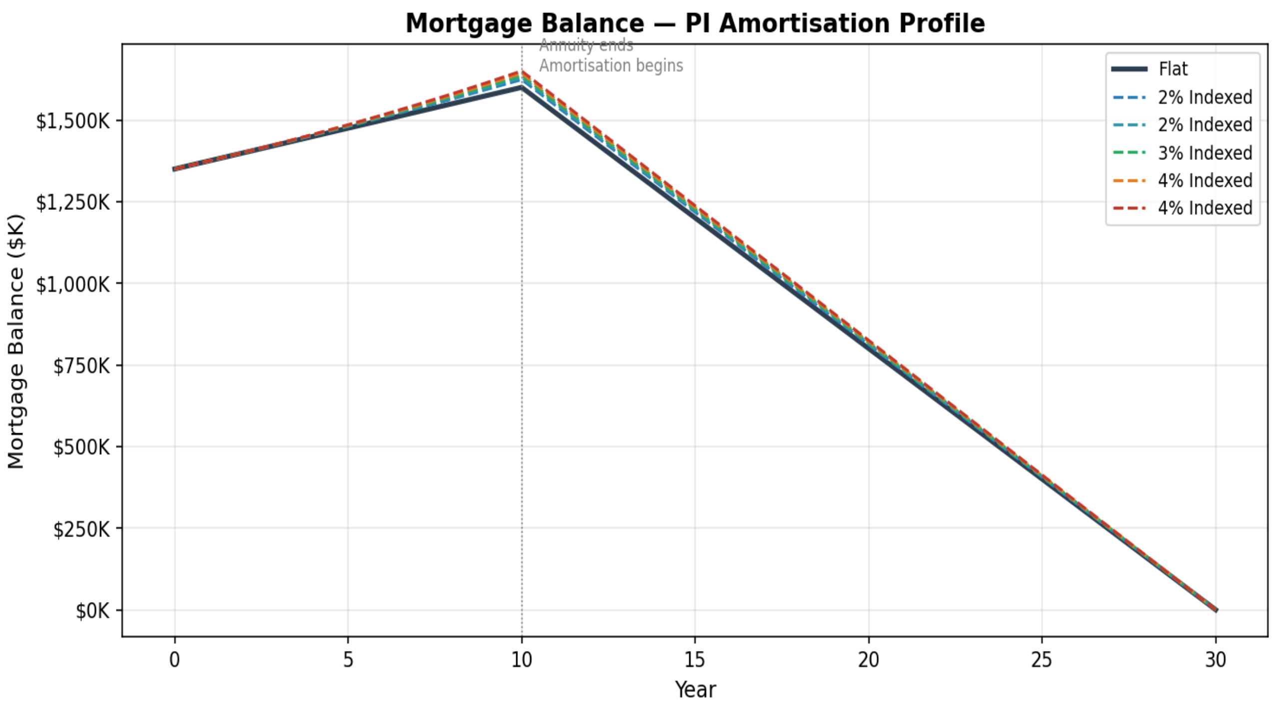
• Insurance premium increases from $12,485 to $15,312 — a 23% increase, but still well below the IO model's $39,541 at the same inflation rate.

2. Optimised Model Structure

The optimised model differs from the baseline in three key respects that collectively reduce PoC from ~14% to ~5%:

- **Principal & Interest Amortisation** — After the 10-year annuity period, the mortgage amortises linearly to $0 over the remaining 20 years. This eliminates the end-of-term balance that drives deficit risk in the IO model. Annual principal repayment: $80,000/year from the investment account.
- **Higher Holiday Entry (1.05x vs 0.90x)** — The holiday mechanism triggers earlier, providing more protection when investment values decline. This increases holiday usage but prevents investment depletion in adverse scenarios.
- **More Frequent Profit Share (20% every 3 years vs 25% every 5 years)** — Smaller, more frequent profit extractions smooth the revenue stream while maintaining total revenue: mean total profit share is $1,287,638.

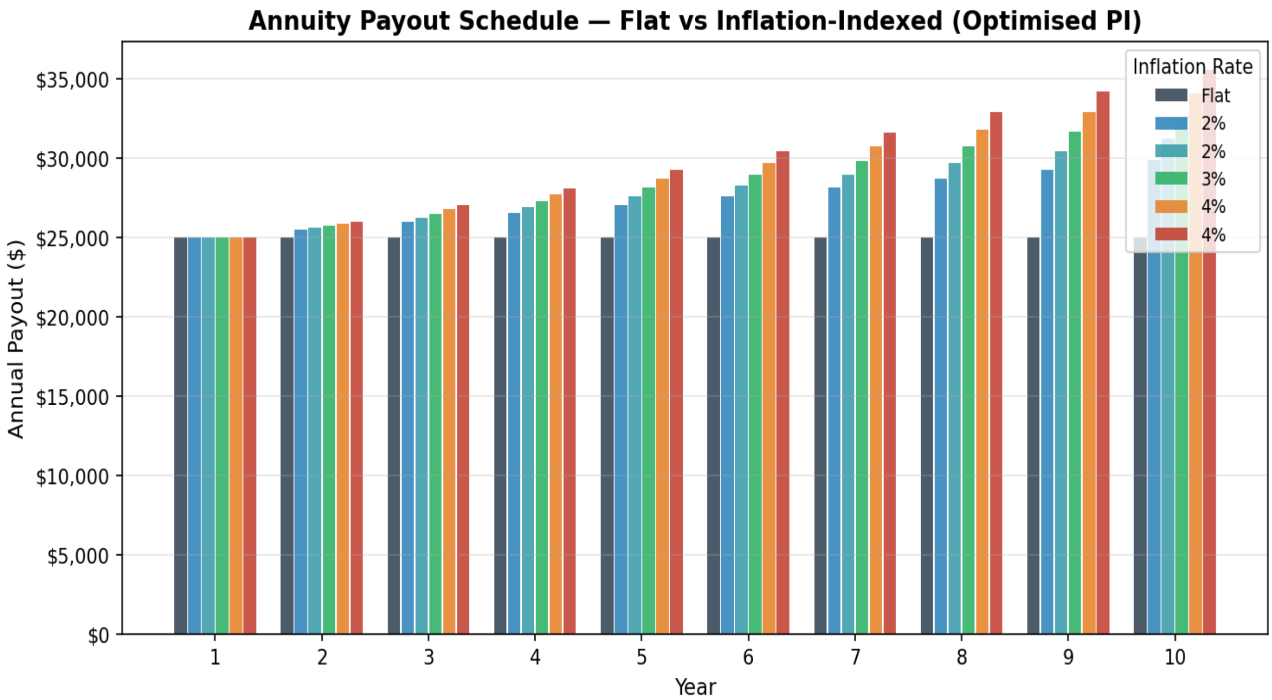
Mortgage Amortisation Profile



Year	Flat	2% Indexed	2% Indexed	3% Indexed	4% Indexed	4% Indexed
Year 0	$1,350,000	$1,350,000	$1,350,000	$1,350,000	$1,350,000	$1,350,000
Year 5	$1,475,000	$1,480,101	$1,481,408	$1,482,728	$1,484,062	$1,485,408
Year 10	$1,600,000	$1,623,743	$1,630,085	$1,636,597	$1,643,285	$1,650,153
Year 15	$1,200,000	$1,217,807	$1,222,563	$1,227,448	$1,232,464	$1,237,615
Year 20	$800,000	$811,872	$815,042	$818,298	$821,642	$825,076
Year 25	$400,000	$405,936	$407,521	$409,149	$410,821	$412,538
Year 30	$0	$0	$0	$0	$0	$0

3. Annuity Payout Schedule

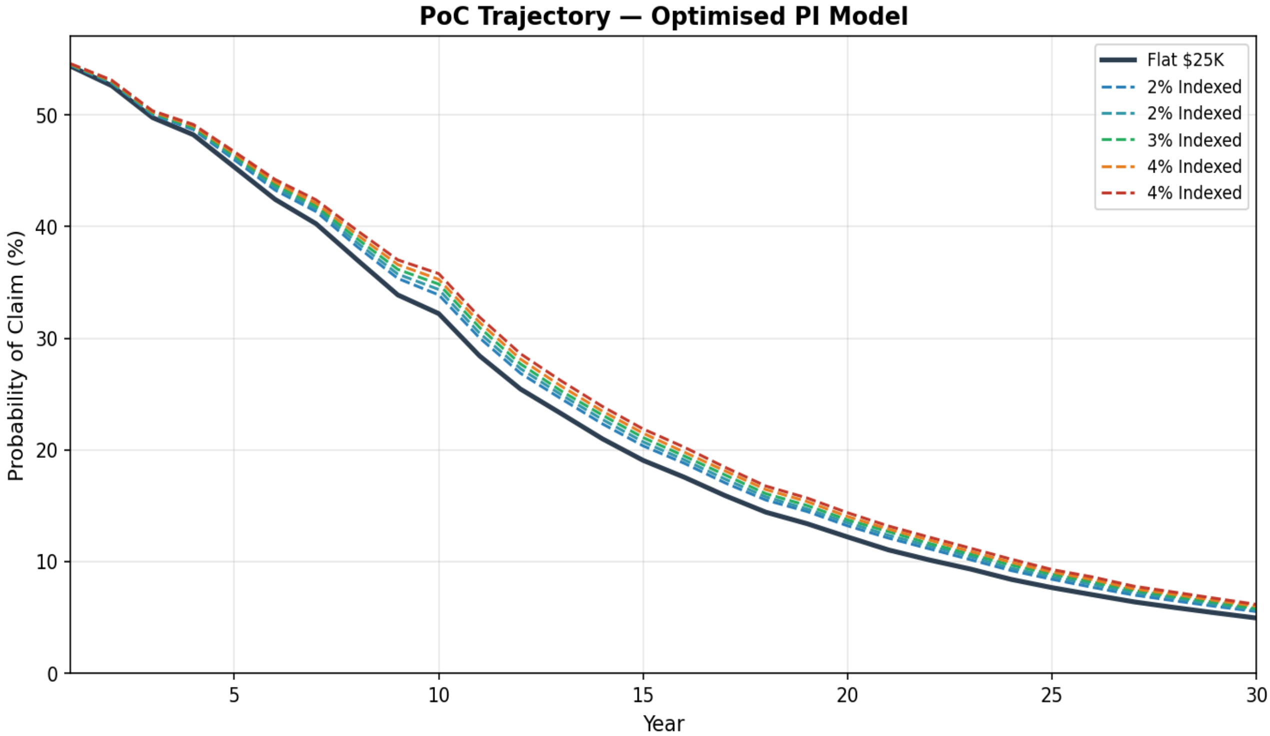
The annuity payout schedule is identical to the IO analysis — the only difference is how the resulting mortgage balance is treated after Year 10.



Year	Flat	2% Indexed	2% Indexed	3% Indexed	4% Indexed	4% Indexed
Year 1	$25,000	$25,000	$25,000	$25,000	$25,000	$25,000
Year 2	$25,000	$25,500	$25,625	$25,750	$25,875	$26,000
Year 3	$25,000	$26,010	$26,266	$26,522	$26,781	$27,040
Year 5	$25,000	$27,061	$27,595	$28,138	$28,688	$29,246
Year 7	$25,000	$28,154	$28,992	$29,851	$30,731	$31,633
Year 10	$25,000	$29,877	$31,222	$32,619	$34,072	$35,583
Total	$250,000	$273,743	$280,085	$286,597	$293,285	$300,153
Extra vs Flat	—	$23,743	$30,085	$36,597	$43,285	$50,153

4. Risk Impact Analysis

4.1 PoC Trajectory

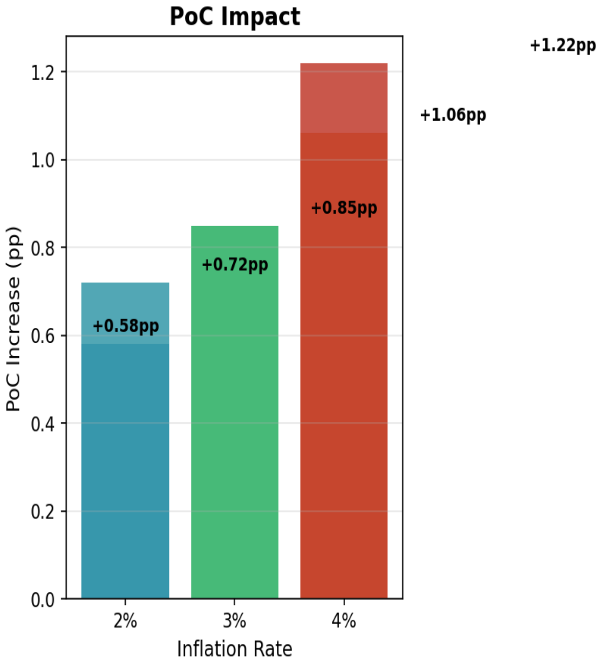


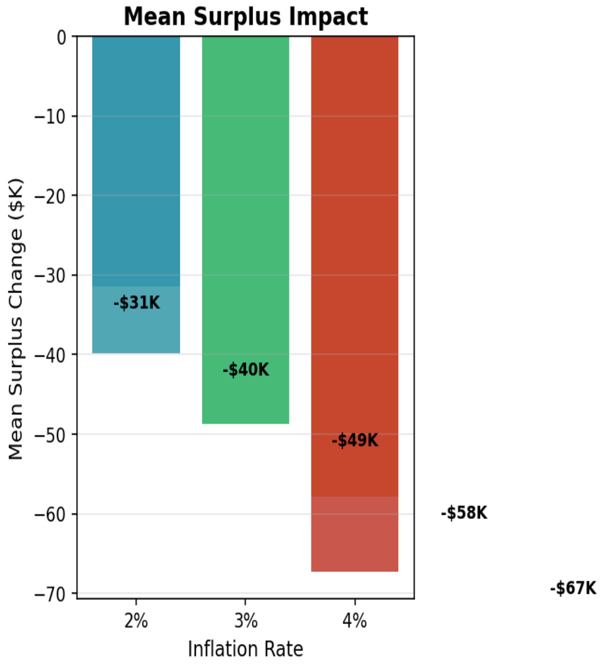
The PoC curves show the hallmark of PI amortisation: a steep decline from Year 10 onwards as principal repayments reduce the mortgage balance. The inflation-indexed scenarios maintain the same shape but with a modest vertical offset. By Year 25, even the 4% scenario (9.28%) is below the flat IO baseline.

4.2 PoC by Year — All Scenarios

Year	Flat	2%	2%	3%	4%	4%
Year 1	54.37%	54.45%	54.48%	54.52%	54.55%	54.59%
Year 5	45.34%	45.99%	46.18%	46.36%	46.54%	46.72%
Year 10	32.21%	33.89%	34.37%	34.85%	35.28%	35.79%
Year 15	19.06%	20.36%	20.72%	21.10%	21.49%	21.87%
Year 20	12.20%	13.20%	13.48%	13.73%	14.03%	14.37%
Year 25	7.66%	8.41%	8.63%	8.87%	9.06%	9.28%
Year 30	4.94%	5.52%	5.66%	5.79%	6.00%	6.16%

5. Surplus Distribution Impact

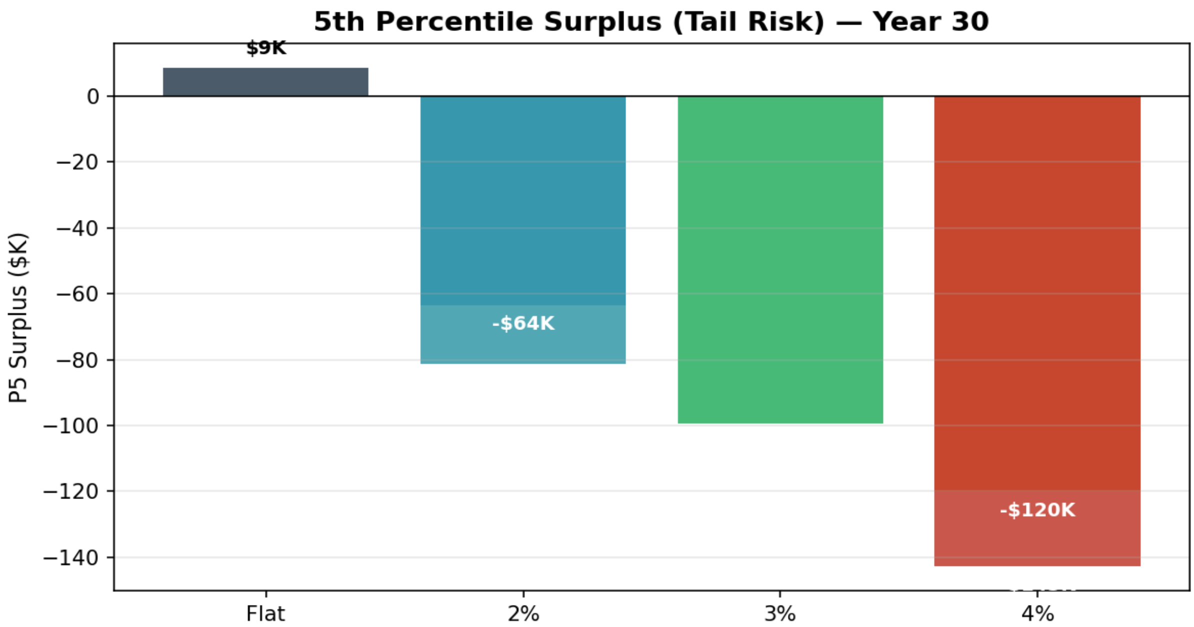




5.1 Key Metrics — All Scenarios (Year 30)

Metric	Flat	2% Indexed	2% Indexed	3% Indexed	4% Indexed	4% Indexed
PoC	4.94%	5.52%	5.66%	5.79%	6.00%	6.16%
Mean Surplus	$1,439,999	$1,408,575	$1,400,166	$1,391,299	$1,382,140	$1,372,682
Median Surplus	$1,299,381	$1,275,526	$1,269,389	$1,262,250	$1,254,916	$1,247,615
P1 Surplus	-$1,037,239	-$1,123,440	-$1,149,359	-$1,179,966	-$1,206,987	-$1,235,289
P5 Surplus	$8,572	-$63,512	-$81,403	-$99,528	-$120,282	-$142,691
P10 Surplus	$353,337	$317,675	$306,325	$297,210	$285,227	$273,420
P25 Surplus	$788,430	$763,161	$756,451	$748,706	$741,273	$734,150
P75 Surplus	$1,960,994	$1,932,863	$1,924,973	$1,916,648	$1,908,267	$1,899,742
P90 Surplus	$2,751,911	$2,720,361	$2,711,612	$2,702,034	$2,693,135	$2,684,474
P95 Surplus	$3,362,649	$3,328,376	$3,318,132	$3,308,769	$3,299,143	$3,289,944
P99 Surplus	$4,784,837	$4,746,758	$4,735,946	$4,726,953	$4,714,224	$4,702,188

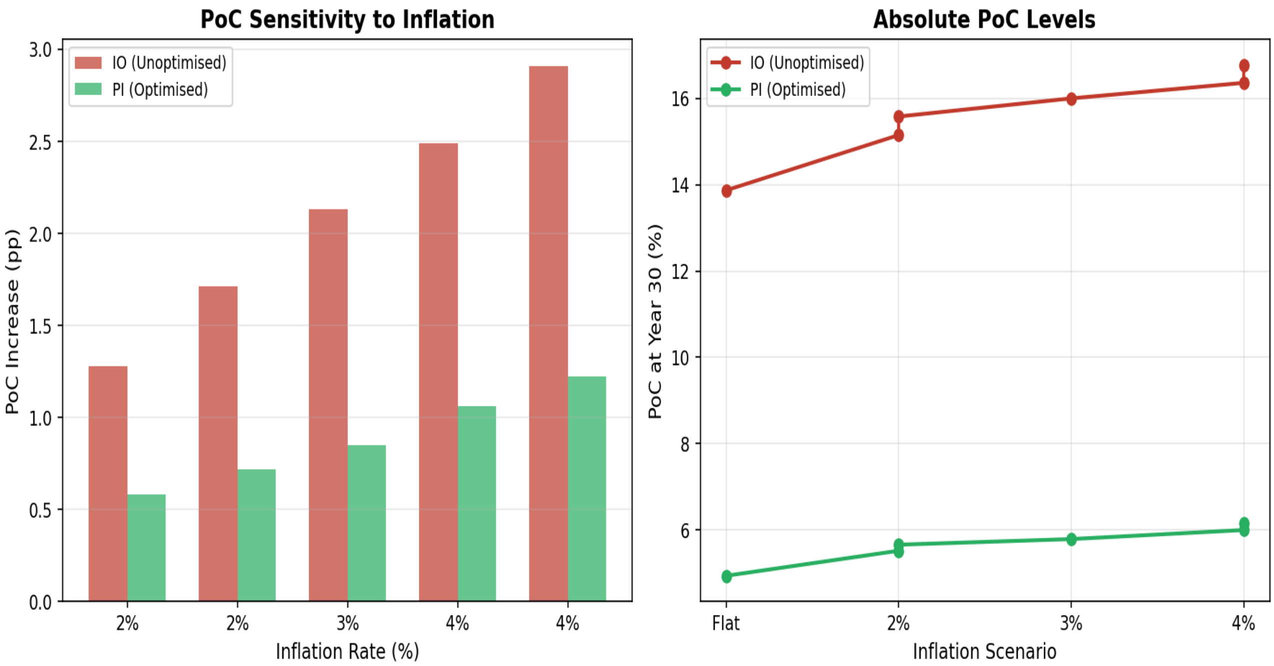
5.2 Tail Risk — 5th Percentile Surplus



A critical observation: the flat PI baseline has a **positive** P5 surplus ($8,572), meaning 95% of paths end in surplus. Inflation indexing at 3%+ pushes P5 negative, but the magnitude is modest relative to IO.

6. IO vs PI Model Comparison

The PI (optimised) model is substantially less sensitive to inflation indexing than the IO (unoptimised) model. This is because PI amortisation reduces the mortgage balance after Year 10, limiting the period over which higher balances compound costs.



Metric	IO Flat	IO @ 3%	IO Δ	PI Flat	PI @ 3%	PI Δ
PoC (Yr 30)	13.87%	16.00%	+2.13pp	4.94%	5.79%	+0.85pp
Mean Surplus	$1,690,289	$1,609,426	-$80,863	$1,439,999	$1,391,299	-$48,700
P5 Surplus	-$653,596	-$768,329	-$114,733	$8,572	-$99,528	-$108,100
P10 Surplus	-$211,973	-$317,540	-$105,567	$353,337	$297,210	-$56,127
Insurance (loaded)	$32,929	$39,541	$6,612	$12,485	$15,312	$2,827
Profit Share	$783,280	$742,937	-$40,343	$1,287,638	$1,235,782	-$51,856

The PI model reduces inflation sensitivity by approximately 60%: PoC increases by +0.85pp vs +2.13pp in IO, and mean surplus declines by $48,700 vs $81K. This makes the optimised model a strong candidate for offering inflation-indexed payouts as a product feature.

7. Cost Multiplier Analysis

The cost multiplier measures how much mean surplus is lost per additional dollar of annuity paid. In the PI model, this is significantly lower than IO because the higher mortgage balance amortises away, limiting the compounding period.

Inflation	Extra Annuity	Surplus Loss	Multiplier	IO Multiplier
2.0%	$23,743	$31,424	1.32x	2.19x
2.5%	$30,085	$39,833	1.32x	2.20x
3.0%	$36,597	$48,700	1.33x	2.21x
3.5%	$43,285	$57,859	1.34x	2.21x
4.0%	$50,153	$67,317	1.34x	2.21x

The PI cost multiplier of ~1.3x means that each additional $1 of inflation-indexed annuity reduces surplus by approximately $1.30 — much closer to a 1:1 relationship than the IO model's ~2.2x. The fundamental reason is that PI amortisation eliminates the mortgage balance by Year 30, so the higher balance from indexed payouts only compounds costs for ~20 years (the amortisation period) rather than the full 30.

8. Conclusions & Recommendations

The optimised PI model provides substantial headroom for inflation-indexed annuity payouts. The structural advantages of principal amortisation absorb the additional cost of indexed payouts with minimal impact on risk metrics.

Assessment

- **2.0–2.5% Inflation** — PoC increases by +0.58pp to +0.72pp, remaining below 6%. P5 surplus moves slightly negative but P10 stays well positive. **Highly recommended** — negligible risk impact with meaningful borrower benefit.
- **3.0% Inflation** — PoC increases by +0.85pp to 5.79%. Mean surplus declines by $48,700. Insurance premium increases by $2,827. **Recommended** — comfortably within risk tolerance.
- **3.5–4.0% Inflation** — PoC reaches 6.16%. Still well below the unoptimised IO baseline of 13.87%. P5 surplus of -$142,691 is the main concern. **Acceptable** — but consider a slight reduction in initial annuity to offset.

Recommendation

The optimised PI model can comfortably support inflation-indexed payouts at 2.5–3.0% without any structural changes. At these rates, PoC remains under 6% (vs the IO model's 16%), and all risk metrics stay within acceptable bounds. This represents a compelling product enhancement: borrowers receive inflation-protected income while the mortgage structure ensures the investment account can absorb the additional cost.

Methodology

All scenarios use the v14a optimised parameters (PI amortisation, holiday entry 1.05, profit share 20%/3yr) with 50,000 Monte Carlo paths and identical random number generation (seed=42). The PI amortisation reduces the mortgage linearly from peak (Year 10) to $0 (Year 30), with principal repayments deducted from the investment account. All other mechanics (holiday, collar, GBM returns, OU cash rate) are identical.